

MORE QUERY'S FOR BANK DATABASE

1. Find all the customers who have an account at all the branches located in a specific city (Ex. Delhi).

SELECT DISTINCT d1.Custormername FROM Depositer d1 WHERE NOT EXISTS (SELECT * FROM branch b WHERE b.Branch_city = 'Delhi' AND NOT EXISTS (SELECT * FROM Depositer d2 JOIN BankAccount ba ON d2.Accno = ba.Accno WHERE d2.Custormername = d1.Custormername AND ba.Branch_name = b.Branch_name));

```
62 • insert into BankCustomer values("Nikil","Akbar_Road","Delhi");
63 • insert into BankCustomer values("Ravi","Prithviraj_Road","Delhi");
64
65 • insert into Depositer values("Avinash",1);
66 • insert into Depositer values("Dinesh",2);
67 • insert into Depositer values("Nikil",4);
68 • insert into Depositer values("Ravi",5);
69 • insert into Depositer values("Avinash",8);
70 • insert into Depositer values("Nikil",9);
71 • insert into Depositer values("Dinesh",10);
72 • insert into Depositer values("Nikil",11);
73
74
75 • insert into Loan values(1,"SBI_Chamrajpet",1000);
76 • insert into Loan values(2,"SBI_ResidencyRoad",2000);
77 • insert into Loan values(3,"SBI_ShivajiRoad",3000);
78 • insert into Loan values(4,"SBI_ParliamentRoad",4000);
79 • insert into Loan values(5,"SBI_Jantarmantra",5000);
80 • select * from branch;
81
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

Branch_name	Branch_city	assets
SBI_Chamrajpet	Bangalore	50000
SBI_Jantarmantra	Delhi	20000
SBI_ParliamentRoad	Delhi	10000
SBI_ResidencyRoad	Bangalore	10000
SBI_ShivajiRoad	Delhi	20000

2. Find all customers who have a loan at the bank but do not have an account.

CREATE TABLE dhiksha_bank.Borrower (

Custormername VARCHAR(20),

Loan_number INT, PRIMARY KEY (Custormername, Loan_number),

FOREIGN KEY (Custormername) REFERENCES BankCustomer(Custormername),

FOREIGN KEY (Loan_number) REFERENCES Loan(Loan_number)

);

INSERT INTO Borrower VALUES ('Avinash', 1);

INSERT INTO Borrower VALUES ('Dinesh', 2);

INSERT INTO Borrower VALUES ('Nikil', 4);

INSERT INTO Borrower VALUES ('Ravi', 5);

insert into Borrower values("Avinash",8);

SELECT DISTINCT b.Customername

FROM Borrower b

WHERE b.Customername NOT IN (

SELECT d.Customername

FROM Depositer d

);

The screenshot shows a database management tool interface. The top section displays a list of SQL queries, including several INSERT statements for BankCustomer, Depositer, and Loan tables, and two SELECT statements. The bottom section shows the results of the queries. The first SELECT statement, 'select * from branch;', returns a table with 5 rows and 3 columns: Accno, Branch_name, and Balance. The second SELECT statement, 'select * from BankAccount;', returns a table with 8 rows and 3 columns: Accno, Branch_name, and Balance. The output section shows the execution of the queries, with messages indicating the number of rows returned for each query.

```
62 • insert into BankCustomer values("Nikil","Akbar_Road","Delhi");
63 • insert into BankCustomer values("Ravi","Prithviraj_Road","Delhi");
64
65 • insert into Depositer values("Avinash",1);
66 • insert into Depositer values("Dinesh",2);
67 • insert into Depositer values("Nikil",4);
68 • insert into Depositer values("Ravi",5);
69 • insert into Depositer values("Avinash",8);
70 • insert into Depositer values("Nikil",9);
71 • insert into Depositer values("Dinesh",10);
72 • insert into Depositer values("Nikil",11);
73
74
75 • insert into Loan values(1,"SBI_Chamrajpet",1000);
76 • insert into Loan values(2,"SBI_ResidencyRoad",2000);
77 • insert into Loan values(3,"SBI_ShivajiRoad",3000);
78 • insert into Loan values(4,"SBI_ParliamentRoad",4000);
79 • insert into Loan values(5,"SBI_Jantarmanatar",5000);
80 • select * from branch;
81 • select * from BankAccount;
```

Accno	Branch_name	Balance
1	SBI_Chamrajpet	2000
2	SBI_ResidencyRoad	5000
4	SBI_ParliamentRoad	9000
5	SBI_Jantarmanatar	8000
8	SBI_ResidencyRoad	4000

Accno	Branch_name	Balance
1	SBI_Chamrajpet	2000
2	SBI_ResidencyRoad	5000
4	SBI_ParliamentRoad	9000
5	SBI_Jantarmanatar	8000
8	SBI_ResidencyRoad	4000

#	Time	Action	Message
1	11:21:37	select * from branch LIMIT 0, 1000	5 row(s) returned
2	11:22:12	select * from BankAccount LIMIT 0, 1000	8 row(s) returned
3	11:22:46	select * from BankAccount LIMIT 0, 1000	8 row(s) returned

3.Find the names of all branches that have greater assets than all branches located in Bangalore.

INSERT INTO BankCustomer VALUES ('Mohan', 'MG Road', 'Bangalore');

INSERT INTO Loan VALUES (6, 'SBI_Chamrajpet', 7000);

INSERT INTO Borrower VALUES ('Mohan', 6);

SELECT DISTINCT b.Customername

FROM Borrower b

WHERE b.Customername NOT IN (

SELECT d.Customername

FROM Depositer d

);

```
64
65 • insert into Depositer values("Avinash",1);
66 • insert into Depositer values("Dinesh",2);
67 • insert into Depositer values("Nikil",4);
68 • insert into Depositer values("Ravi",5);
69 • insert into Depositer values("Avinash",8);
70 • insert into Depositer values("Nikil",9);
71 • insert into Depositer values("Dinesh",10);
72 • insert into Depositer values("Nikil",11);
73
74
75 • insert into Loan values(1,"SBI_Channarayana",1000);
76 • insert into Loan values(2,"SBI_ResidencyRoad",2000);
77 • insert into Loan values(3,"SBI_ShivajiRoad",3000);
78 • insert into Loan values(4,"SBI_ParliamentRoad",4000);
79 • insert into Loan values(5,"SBI_JantaMandir",5000);
80 • select * from branch;
81 • select * from BankAccount;
82 • select * from BankCustomer;
83
```

Result Grid

Customername	Customer_street	CustomerCity
Avinash	Bull_Temple_Road	Bangalore
Dinesh	Bannergatta_Road	Bangalore
Mohan	NationalCollege_Road	Bangalore
Nikil	Alkbar_Road	Delhi
Ravi	Prithviraj_Road	Delhi

ankCustomer 17 x Apply

Output

#	Time	Action	Message
1	11:21:37	select * from branch LIMIT 0, 1000	5 row(s) returned
2	11:22:12	select * from BankAccount LIMIT 0, 1000	8 row(s) returned
3	11:22:46	select * from BankAccount LIMIT 0, 1000	8 row(s) returned
4	11:23:20	select * from BankCustomer LIMIT 0, 1000	5 row(s) returned

4.Find all customers who have both an account and a loan at the Bangalore branch.

SELECT Branch_name

FROM branch

WHERE assets > ALL (

SELECT assets

FROM branch

WHERE Branch_city = 'Bangalore'

);

INSERT INTO branch VALUES ('SBI_MumbaiCentral', 'Mumbai', 60000);

The screenshot displays a database management interface with a SQL editor and a results pane. The SQL editor contains the following queries:

```
65 • insert into Depositer values("Avinash",1);
66 • insert into Depositer values("Dinesh",2);
67 • insert into Depositer values("Nikil",4);
68 • insert into Depositer values("Ravi",5);
69 • insert into Depositer values("Avinash",8);
70 • insert into Depositer values("Nikil",9);
71 • insert into Depositer values("Dinesh",10);
72 • insert into Depositer values("Nikil",11);
73
74
75 • insert into Loan values(1,"SBI_Chamrajpet",1000);
76 • insert into Loan values(2,"SBI_ResidencyRoad",2000);
77 • insert into Loan values(3,"SBI_ShivajiRoad",3000);
78 • insert into Loan values(4,"SBI_ParliamentRoad",4000);
79 • insert into Loan values(5,"SBI_Jantar Mantar",5000);
80 • select * from branch;
81 • select * from BankAccount;
82 • select * from BankCustomer;
83 • select * from Depositer;
```

The results pane shows the output of the first query, which is a table with two columns: Customername and Accno. The data is as follows:

Customername	Accno
Avinash	1
Dinesh	2
Nikil	4
Ravi	5
Avinash	8

Below the table, there is an 'Output' section showing the execution of the queries. The output is as follows:

#	Time	Action	Message
1	11:21:37	select * from branch LIMIT 0, 1000	5 row(s) returned
2	11:22:12	select * from BankAccount LIMIT 0, 1000	8 row(s) returned
3	11:22:46	select * from BankCustomer LIMIT 0, 1000	8 row(s) returned
4	11:23:20	select * from BankCustomer LIMIT 0, 1000	5 row(s) returned
5	11:23:46	select * from Depositer LIMIT 0, 1000	8 row(s) returned

5.Demonstrate how you delete all account tuples at every branch located in a specific city (Ex. Bomay).

SELECT DISTINCT br.Customername

FROM Borrower br

JOIN Loan l ON br.Loan_number = l.Loan_number

WHERE l.Branch_name IN (

SELECT Branch_name

FROM branch

WHERE Branch_city = 'Bangalore'

)

AND br.Customername IN (

SELECT d.Customername

```

FROM Depositer d

JOIN BankAccount ba ON d.Accno = ba.Accno

WHERE ba.Branch_name IN (

    SELECT Branch_name

    FROM branch

    WHERE Branch_city = 'Bangalore'

)

);

```

The screenshot displays a database management interface with a SQL editor and a results pane. The SQL editor contains the following queries:

```

66 insert into Depositer values('Dinesh',2);
67 insert into Depositer values('Wikil',4);
68 insert into Depositer values('Ravi',5);
69 insert into Depositer values('Avinash',8);
70 insert into Depositer values('Wikil',9);
71 insert into Depositer values('Dinesh',10);
72 insert into Depositer values('Wikil',11);
73
74
75 insert into Loan values(1,'SBI_Chamrajpet',1000);
76 insert into Loan values(2,'SBI_ResidencyRoad',2000);
77 insert into Loan values(3,'SBI_ShivajiRoad',3000);
78 insert into Loan values(4,'SBI_ParliamentRoad',4000);
79 insert into Loan values(5,'SBI_Jantarmantra',5000);
80 select * from branch;
81 select * from BankAccount;
82 select * from BankCustomer;
83 select * from Depositer;
84 select * from Loan;
85

```

The results pane shows the output of the queries. The first query, 'select * from branch', returns 5 rows:

Loan_number	Branch_name	Amount
1	SBI_Chamrajpet	1000
2	SBI_ResidencyRoad	2000
4	SBI_ParliamentRoad	4000
5	SBI_Jantarmantra	5000
3	SBI_ShivajiRoad	3000

The second query, 'select * from BankAccount', returns 8 rows. The third query, 'select * from BankCustomer', returns 5 rows. The fourth query, 'select * from Depositer', returns 8 rows. The fifth query, 'select * from Loan', returns 4 rows.

6.Update the Balance of all accounts by 5%.

```
DELETE FROM BankAccount
```

```
WHERE Branch_name IN (
```

```
    SELECT Branch_name
```

```
    FROM branch
```

```
    WHERE Branch_city = 'Bombay'
```

);

select * from BankAccount;

The screenshot shows a SQL IDE with a script editor and a result grid. The script contains the following SQL statements:

```
66 • insert into Depositer values("Dinesh",2);
67 • insert into Depositer values("Nikil",4);
68 • insert into Depositer values("Ravi",5);
69 • insert into Depositer values("Avinash",8);
70 • insert into Depositer values("Nikil",9);
71 • insert into Depositer values("Dinesh",10);
72 • insert into Depositer values("Nikil",11);
73
74
75 • insert into Loan values(1,"SBI_Chamrajpet",1000);
76 • insert into Loan values(2,"SBI_ResidencyRoad",2000);
77 • insert into Loan values(3,"SBI_ShivajiRoad",3000);
78 • insert into Loan values(4,"SBI_ParliamentRoad",4000);
79 • insert into Loan values(5,"SBI_Jantarmanatar",5000);
80 • select * from branch;
81 • select * from BankAccount;
82 • select * from BankCustomer;
83 • select * from Depositer;
84 • select * from Loan;
85 • select Branch_name, CONCAT(assets/100000," lakhs')assets_in_lakhs from branch;
```

The result grid shows the output of the last statement, displaying a table with two columns: Branch_name and assets_in_lakhs.

Branch_name	assets_in_lakhs
SBI_Chamrajpet	0.5000 lakhs
SBI_Jantarmanatar	0.2000 lakhs
SBI_ParliamentRoad	0.1000 lakhs
SBI_ResidencyRoad	0.1000 lakhs
SBI_ShivajiRoadBombay	0.2000 lakhs

Below the result grid, there is an 'Action Output' section showing a log of SQL actions and their results.

#	Time	Action	Message
1	11:21:37	select * from branch LIMIT 0, 1000	5 row(s) returned
2	11:22:12	select * from BankAccount LIMIT 0, 1000	8 row(s) returned
3	11:22:46	select * from BankCustomer LIMIT 0, 1000	8 row(s) returned
4	11:23:20	select * from Depositer LIMIT 0, 1000	5 row(s) returned
5	11:23:46	select * from Loan LIMIT 0, 1000	8 row(s) returned
6	11:24:24	select * from Loan LIMIT 0, 1000	4 row(s) returned
7	11:25:23	select Branch_name, CONCAT(assets/100000," lakhs')assets_in_lakhs from branch LIMIT 0, 1000	5 row(s) returned

7.LIST THE ENTIRE LOAN RELATION IN THE DESCENDING ORDER OF AMOUNT.

UPDATE BankAccount

SET Balance = ROUND(Balance * 1.05);

SELECT Accno, Balance, ROUND(Balance * 1.05) AS NewBalance

FROM BankAccount;

68 • insert into Depositer values("Ravi",5);
69 • insert into Depositer values("Nikil",9);
70 • insert into Depositer values("Dinesh",10);
71 • insert into Depositer values("Nikil",11);
72 •
73
74
75 • insert into Loan values(1,"SBI_Chamrajpet",1000);
76 • insert into Loan values(2,"SBI_ResidencyRoad",2000);
77 • insert into Loan values(3,"SBI_ShivajiRoad",3000);
78 • insert into Loan values(4,"SBI_ParliamentRoad",4000);
79 • insert into Loan values(5,"SBI_Jantarmantra",5000);
80 • select * from branch;
81 • select * from BankAccount;
82 • select * from BankCustomer;
83 • select * from Depositer;
84 • select * from Loan;
85 • select Branch_name, CONCAT(assets/100000,' lakhs')assets_in_lakhs from branch;
86 • select d.Customername from Depositer d, BankAccount b where b.Branch_name="SBI_ResidencyRoad" and d.Accno=b.Accno group by d.Customername having count(d.Accno)>=2;
87

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

Customername
Dinesh

Result 21 x Read O

Output:

#	Time	Action	Message
1	11:21:37	select * from branch LIMIT 0, 1000	5 row(s) returned
2	11:22:12	select * from BankAccount LIMIT 0, 1000	8 row(s) returned
3	11:22:46	select * from BankCustomer LIMIT 0, 1000	8 row(s) returned
4	11:23:20	select * from BankCustomer LIMIT 0, 1000	5 row(s) returned
5	11:23:46	select * from Depositer LIMIT 0, 1000	8 row(s) returned
6	11:24:24	select * from Loan LIMIT 0, 1000	4 row(s) returned
7	11:25:23	select Branch_name, CONCAT(assets/100000,' lakhs')assets_in_lakhs from branch LIMIT 0, 1000	5 row(s) returned
8	11:25:59	select d.Customername from Depositer d, BankAccount b where b.Branch_name="SBI_ResidencyRoad" and d.Accno=b.Accno group by d.Customername having count(d.Accno)>=2;	1 row(s) returned

8.FIND ALL CUSTOMERS HAVING A LAON, AN ACCOUNT OR BOTH AT THE BANK.

SELECT *

FROM Loan

ORDER BY Amount DESC;

SELECT Customername FROM Depositer

UNION

SELECT Customername FROM Borrower;

```

72 • insert into Depositer values("nikil",11);
73
74
75 • insert into Loan values(1,"SBI_Channajpet",1000);
76 • insert into Loan values(2,"SBI_ResidencyRoad",2000);
77 • insert into Loan values(3,"SBI_ShivajiRoad",3000);
78 • insert into Loan values(4,"SBI_ParliamentRoad",4000);
79 • insert into Loan values(5,"SBI_Jantaramantar",5000);
80 • select * from branch;
81 • select * from BankAccount;
82 • select * from BankCustomer;
83 • select * from Depositer;
84 • select * from Loan;
85 • select Branch_name, CONCAT(assets/100000,' lakhs')assets_in_lakhs from branch;
86 • select d.Customername from Depositer d, BankAccount b where b.Branch_name='SBI_ResidencyRoad' and d.Accno=b.Accno group by d.Customername having count(d.Accno)>=2;
87 • create view sum_of_loan
88 as select Branch_name, SUM(Balance)
89 from BankAccount
90 group by Branch_name;
91 • select * from sum_of_loan;

```

Branch_name	SUM(Balance)
SBI_Channajpet	2000
SBI_Jantaramantar	10000
SBI_ParliamentRoad	12000
SBI_ResidencyRoad	14000

sum_of_loan 22 x

Output

9.CREATE A VIEW WHICH GIVES EACH BRANCH THE SUM OF THE AMOUNT OF ALL THE LOANS AT THE BRANCH.

CREATE VIEW Branch_Total_Loan AS

SELECT Branch_name, SUM(Amount) AS Total_Loan

FROM Loan

GROUP BY Branch_name;

```

81 • select * from BankAccount;
82 • select * from BankCustomer;
83 • select * from Depositer;
84 • select * from Loan;
85 • select Branch_name, CONCAT(assets/100000,' lakhs')assets_in_lakhs from branch;
86 • select d.Customername from Depositer d, BankAccount b where b.Branch_name='SBI_ResidencyRoad' and d.Accno=b.Accno group by d.Customername having count(d.Accno)>=2;
87 • create view sum_of_loan
88 as select Branch_name, SUM(Balance)
89 from BankAccount
90 group by Branch_name;
91 • select * from sum_of_loan;
92 • select bc.Customername, CONCAT(Balance*1000,' rupees') UPDATED_BALANCE from BankAccount b, BankCustomer bc, Depositer d
93 where bc.Customername=d.Customername and b.Accno=d.Accno and bc.Customercity='Bangalore';
94
95
96
97
98
99
100

```

Customername	UPDATED_BALANCE
Avinash	3000 rupees
Avinash	5000 rupees
Dinesh	6000 rupees
Dinesh	6000 rupees

Result 23 x

Output

Action Output

#	Time	Action	Message
1	11:28:27	select bc.Customername, CONCAT(Balance*1000,' rupees') UPDATED_BALANCE from BankAccount b, BankCustomer bc, Depositer d where bc.Cust...	4 row(s) returned

10.THE ANNUAL INTEREST PAYMENTS ARE MADE AND ALL BRANCHES ARE TO BE INCREASED

BY 5%.

SELECT Accno, Balance, ROUND(Balance * 1.05) AS NewBalance

FROM BankAccount;

The screenshot shows a SQL IDE interface. The top toolbar includes icons for file operations, execution, and a 'Limit to 1000 rows' dropdown. The script editor contains the following SQL statements:

```
66 • insert into Depositer values("Dinesh",2);
67 • insert into Depositer values("Nikil",4);
68 • insert into Depositer values("Ravi",5);
69 • insert into Depositer values("Avinash",8);
70 • insert into Depositer values("Nikil",9);
71 • insert into Depositer values("Dinesh",10);
72 • insert into Depositer values("Nikil",11);
73
74
75 • insert into Loan values(1,"SBI_Chamrajpet",1000);
76 • insert into Loan values(2,"SBI_ResidencyRoad",2000);
77 • insert into Loan values(3,"SBI_ShivajiRoad",3000);
78 • insert into Loan values(4,"SBI_ParliamentRoad",4000);
79 • insert into Loan values(5,"SBI_Jantarmanatar",5000);
80 • select * from branch;
81 • select * from BankAccount;
82 • select * from BankCustomer;
83 • select * from Depositer;
84 • select * from Loan;
85 • select Branch_name, CONCAT(assets/100000," lakhs')assets_in_lakhs from branch;
```

Below the script editor is a 'Result Grid' with columns 'Branch_name' and 'assets_in_lakhs'. It displays the following data:

Branch_name	assets_in_lakhs
SBI_Chamrajpet	0.5000 lakhs
SBI_Jantarmanatar	0.2000 lakhs
SBI_ParliamentRoad	0.1000 lakhs
SBI_ResidencyRoad	0.1000 lakhs
SBI_ShivajiRoadBombay	0.2000 lakhs

Below the result grid is an 'Output' section with a table showing the execution of the script:

#	Time	Action	Message
1	11:21:37	select * from branch LIMIT 0, 1000	5 row(s) returned
2	11:22:12	select * from BankAccount LIMIT 0, 1000	8 row(s) returned
3	11:22:46	select * from BankCustomer LIMIT 0, 1000	8 row(s) returned
4	11:23:20	select * from Depositer LIMIT 0, 1000	5 row(s) returned
5	11:23:46	select * from Loan LIMIT 0, 1000	8 row(s) returned
6	11:24:24	select * from Loan LIMIT 0, 1000	4 row(s) returned
7	11:25:23	select Branch_name, CONCAT(assets/100000,' lakhs')assets_in_lakhs from branch LIMIT 0, 1000	5 row(s) returned